
Security Login Analytics Investigation

Short Technical Review

All key decisions, anomaly evidence, product analytics,
and production design in one compact deck.

Prepared by Stasy | 10 158 login events | 2024-01-01 to 2024-03-29



Executive summary

The investigation identifies repeatable suspicious behaviors and a path to operationalize them.

Events analyzed

10 158

100 users

Dataset window

89 days

2024-01-01 to 2024-03-29

Overall failure rate

6.2%

634 failures

Sharpest spike

Feb 9

37.1% failure rate

Priority finding	Confidence	Why it matters
Brute-force burst	95%	Machine-speed timing on user_097
Shared-IP spray	92%	20 users from one IP at one timestamp
Nightly probe	90%	Repeated exact 20:00 failures before success
IP velocity	70%	Strong signal, but threshold needs calibration

The strongest answer is a repeatable framework, not a collection of isolated suspicious rows.

Assignment scope balances security and product analytics

The work treats attack reduction and business growth as equal analytical responsibilities.



Question	Security answer	Product answer
What changed?	Failure spikes and suspicious sources	Signup volume and channel mix
Is it good?	Only if false positives are controlled	Only if activation and quality hold
What action follows?	Triage, enrich, contain, learn	Segment, verify, improve conversion

Analytical strategy and decision log

The approach favors explainable behavioral analytics before machine learning.

Decision	Alternatives	Why rejected
Rule-based detection	Isolation Forest	No labels and lower SOC explainability
Daily baseline	Weekly baseline	Dataset too short for stable weekly seasonality
Failure rate	Raw failures	Normalizes activity across users and sources
IP velocity	Raw IP count	Makes source spread comparable across users
Composite score	Single flag	Combines timing, outcome, source, and novelty

Heuristics are appropriate first because the dataset has no labels, limited context, and requires defensible explanations.

Dataset and data quality

Small integrity issues can materially distort security scoring when alerts become automated.

Login records

10 158

Raw authentication events

Duplicates

88

Deduplicate before scoring

Missing IP

3

Weakens source analytics

Missing outcome

3

Affects failure metrics

Issue	Count	Business impact	Action
Missing IP	3	Weakens IP-based detection	Investigate upstream
Missing outcome	3	Affects failure metrics	Conservative handling
Duplicates	88	Inflates alerts	Deduplicate before scoring
Invalid labels	2	Schema inconsistency	Normalize values

Baseline and Feb 9 spike

Aggregate trends identify when to drill into user and IP behavior.

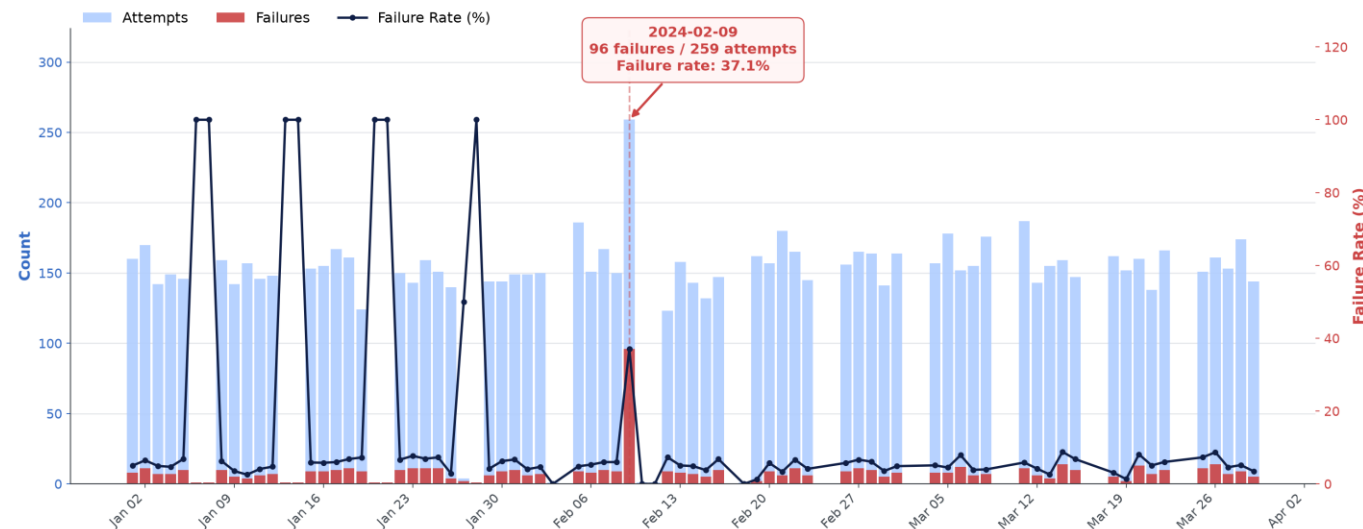
Daily Login Attempts and Failures

Total Attempts
10,158
Avg / day: 152

Total Failures
634
Avg / day: 8.2

Failure Rate
6.24%
Avg daily rate: 14.67%

Date Range
Jan 01 - Mar 29, 2024
77 days



Feb 9 attempts

259

Dataset-level spike

Feb 9 failures

96

Concentrated failures

Failure rate

37.1%

Far above baseline

Feb 9 is not the conclusion; it is the signpost that points to concentrated attacks.

Behavioral features used for scoring

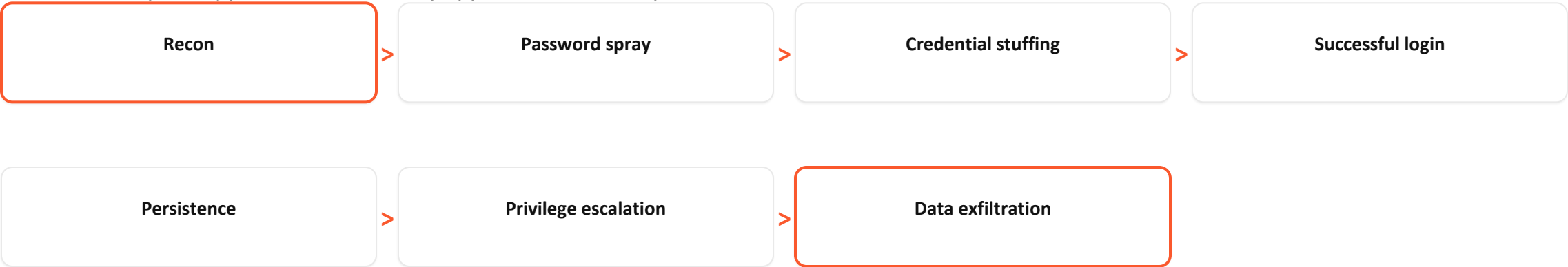
Raw events are converted into comparable user, IP, and timing signals.

Feature	What it captures	Why it is better than raw rows
Failure rate	Failures / attempts	Normalizes activity volume
IP velocity	Distinct IPs / attempts	Detects rotating sources
Attempt velocity	Events in tight windows	Finds automation
Shared-IP spread	Users per IP per timestamp	Finds spraying
Off-hours access	00:00 to 05:00 logins	Adds context when paired with novelty

Each feature is interpretable enough for an analyst to validate without trusting a black box.

Threat kill chain mapping

Each anomaly is mapped to where it likely appears in an attack path.



Anomaly	Kill-chain fit	Analyst action
user_097 burst	Credential stuffing / automation	Block IP, force reset, inspect sessions
user_003 nightly probe	Recon to successful login	Step-up MFA and review post-login activity
Shared-IP spray	Password spray	Open case, enrich IP, look for broad exposure
High IP velocity	Recon / distributed attack	Calibrate with geo, ASN, and device novelty

Anomaly confidence at a glance

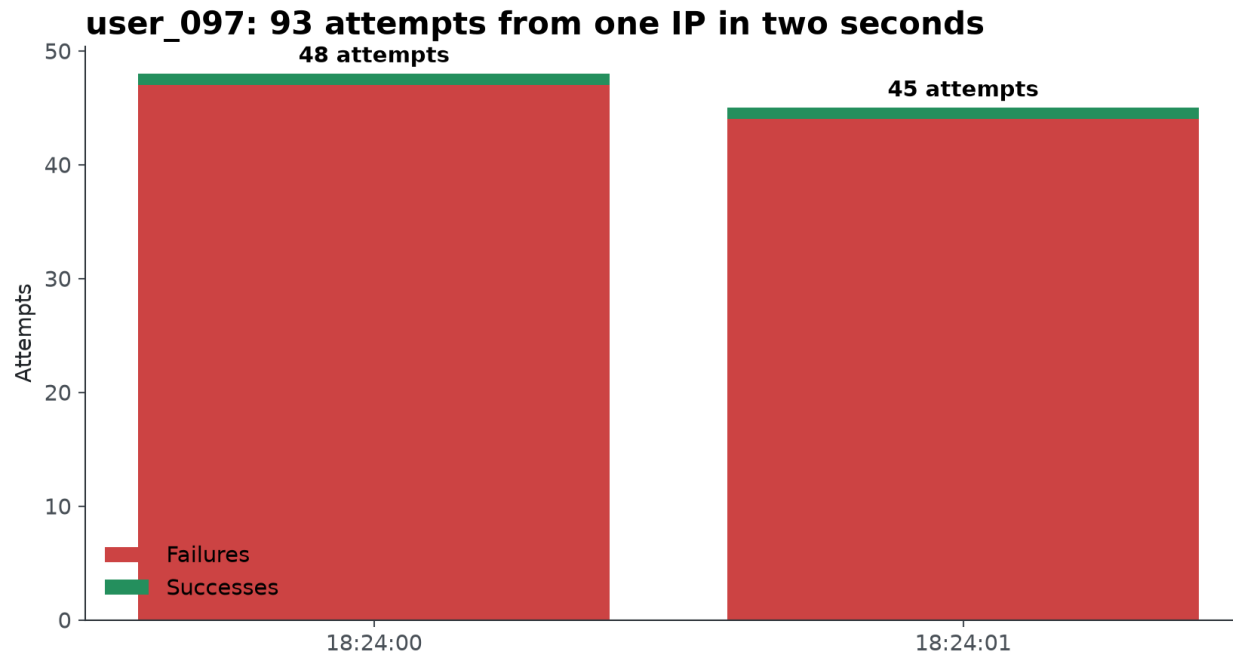
The confidence score summarizes evidence strength, alternatives, and remaining uncertainty.

Case	Confidence	Evidence	Remaining uncertainty
user_097 burst	95%	48 attempts at one second; same IP; repeated failures	Could include replayed logs
Shared-IP spray	92%	20 users, one IP, same timestamp, mixed outcomes	Need IP reputation and ASN
user_003 probe	90%	33 exact 20:00 attempts; 32 failures then success	Need user schedule context
IP velocity	70%	Many users have broad IP spread	Threshold needs calibration
Off-hours	65%	5 midnight logins; user_009 has 4	Work schedule may explain it

Confidence is not certainty; it is a transparent way to prioritize triage.

Anomaly 1: user_097 brute-force burst

Confidence: 95% - impossible timing is the strongest signal.



Evidence

- 48 attempts at 2024-02-09 18:24:00
- Same IP: 103.56.23.138
- 97 failures and 50.0% failure rate

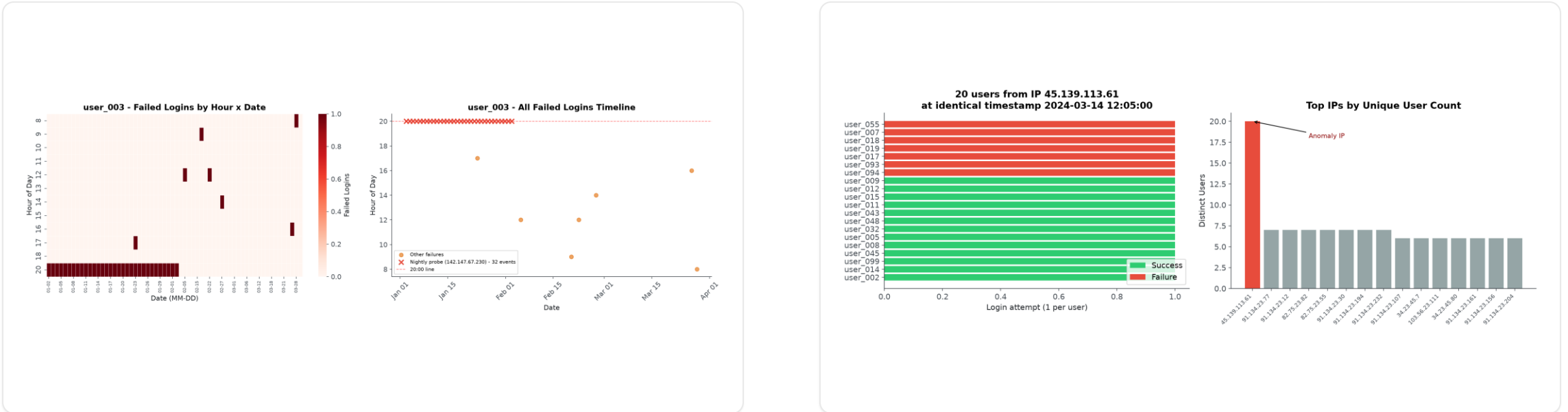
Alternatives

- Duplicate ingestion
- Replay artifacts
- Synthetic load test

Contain first, then verify with session history, source reputation, and upstream log quality.

Anomalies 2 and 3: probe plus shared-IP spray

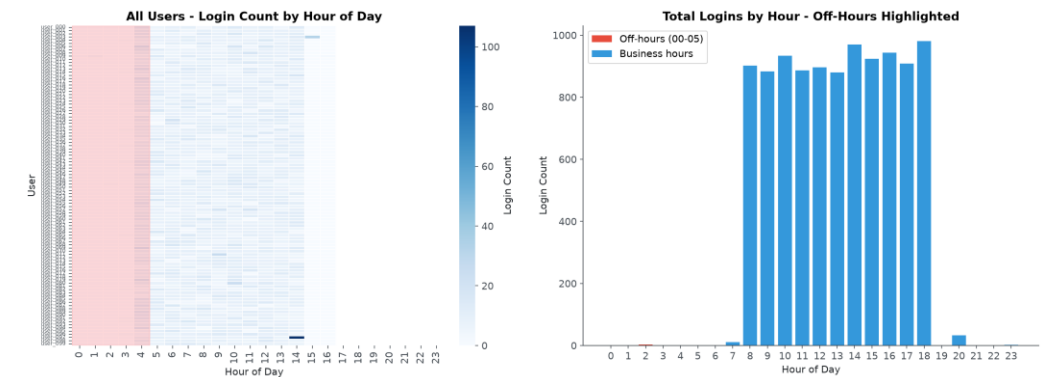
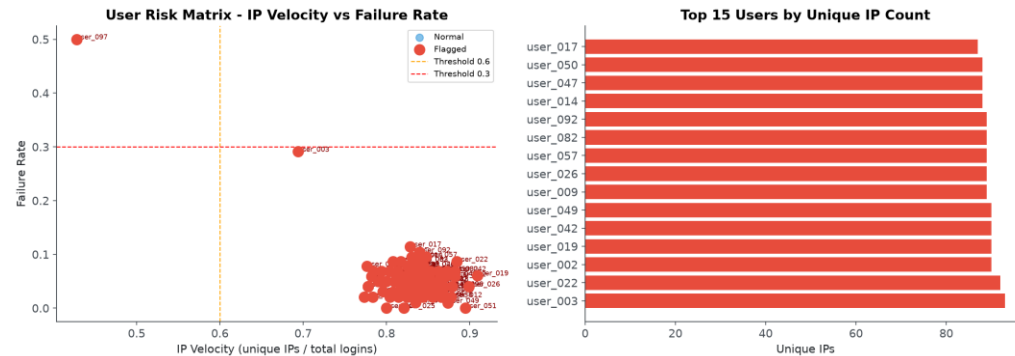
Both patterns indicate coordinated source behavior rather than normal user mistakes.



Case	Confidence	Evidence	Action
user_003	90%	33 exact 20:00 probes; 32 failures; 1 success	Step-up MFA
45.139.113.61	92%	20 users at one timestamp; 13 successes	Open case and enrich IP

Anomalies 4 and 5: calibration-sensitive signals

IP velocity and off-hours access are useful, but should not fire alone.



Signal	Confidence	Interpretation	Production hardening
High IP velocity	70%	Potential proxy/VPN/bot source rotation	Add geo, ASN, device and user history
Off-hours access	65%	Suspicious only with novelty or sensitive actions	Use work schedules and role context

Product KPI case: signup spike needs quality checks

A growth spike is valuable only if acquisition quality and downstream activation hold.

Dashboard questions

- Which channel, country, and device changed?
- Did verification, activation, and payment quality hold?
- Was the spike real traffic, bot abuse, or instrumentation drift?
- What revenue or retention signal followed the spike?

Core KPIs

- Signups
- Activation
- Verified users
- Fraud rate
- Revenue impact

New Customer Sign-Ups - KPI Monitoring Dashboard

Red shading = surge period (Mar 25+) | Quality drop-off during spike



Why explainability and cost matter

Security teams need alerts they can trust, defend, tune, and afford to run.

Dimension	Heuristics first	Machine learning later
SOC trust	Clear evidence per alert	Needs model explanation layer
Legal defensibility	Rule and threshold are inspectable	Requires governance and documentation
Tuning	Fast threshold changes	Needs retraining and drift monitoring
Latency	Milliseconds	Higher inference and feature costs
Maintenance	Low to moderate	Model monitoring, retraining, feature parity

The right path is not anti-ML; it is staged automation after evidence, labels, and feedback exist.

Production architecture expands the notebook into a detection platform

The same features become streaming signals, alert rules, and feedback data.



Analyst feedback closes the loop: it improves thresholds now and creates labels for future ML.

Recommendations and conclusion

The final recommendation is a repeatable, defensible analytics framework.

Time horizon	Action	Owner value
Now	Deduplicate, normalize labels, triage top anomalies	Cleaner alerts and faster response
Next	Add geo, ASN, user agent, MFA, session context	Fewer false positives
Next	Build SOC dashboard with confidence and evidence	Analyst trust and auditability
Later	Use feedback labels for ML and drift monitoring	Gradual automation

Close

This investigation demonstrates a repeatable analytical framework rather than a collection of isolated findings. I prioritized interpretable behavioral analytics because the dataset lacked labels and production context. The proposed features, workflow, dashboard, and architecture support analysts today while creating a path toward a production-grade detection platform.